Chapters 2 and 3, standalone working draft

Restructured skeleton. Prose is provisional, citations are placeholders.

2 Truncation Bias and the Case for Long-Lag VARs

A VAR used for structural inference is exposed to several distinct biases: small sample bias, identification bias (Forni et al., 2016), and truncation bias (De Graeve and Westermark, 2025; Ravenna, 2007; Chari et al., 2008). This chapter is about the third, and about the way it contaminates the second.

The argument runs in three steps. First, the condition usually invoked to licence a VAR, that the model admits an exact infinite order VAR representation, is not the condition the present exercise needs, and abandoning it does not weaken the case for care over the lag order but strengthens it. Second, truncation distorts the estimated responses at every horizon, and it does so in two distinct ways depending on whether the horizon lies inside or outside the estimated lag structure. Third, and as a consequence, the lag order is not a nuisance setting to be delegated to an information criterion. It should be chosen to be consistent with the horizons that the research question, and in impulse response matching the estimator itself, actually use.

2.1 What the VAR is required to deliver

The textbook justification for confronting a DSGE model with a VAR runs through a chain of conditions. The log-linearised solution is written in state space form; if the system is square, so that the number of structural shocks equals the number of observables, and if the Poor Man's Invertibility Condition of Fernández-Villaverde et al. (2007) holds, then the state space system can be inverted into a VAR(∞) in the observables; the researcher then truncates that representation at some finite p. On this reading the VAR is the exact reduced form of the model, and truncation is the only approximation being made.

The chain is a sufficient condition for a particular and rather strong interpretation of the VAR. It is not a necessary condition for the VAR to be useful. Impulse response matching makes a weaker claim. The statement is not "my model generated the data" but "whatever generated the data, the response of these variables to this shock looks like this over these horizons, and my model should reproduce it". Forni et al. (2016) make the point directly: fundamentalness is "extremely and unnecessarily restrictive", because "in several cases fundamentalness is not needed. The researcher is often interested in assessing the transmission mechanisms of a single shock." What matters "is not whether we have fundamentalness or not, but whether we have enough information to get a reasonably good approximation for the specific shocks of interest".

This reframing is liberating, but it should not be misread. Once the VAR is no longer defended as the exact representation of anything, its entire claim on our attention is that it approximates the responses of interest well over the horizons of interest. Truncation is precisely what governs the quality of that approximation. Dropping the VAR(∞) requirement therefore does not make the lag order a free parameter. It concentrates the whole burden of the exercise onto it.

2.2 A simple illustration

Following De Graeve and Westermark (2025), consider a data generating process in which y_t depends on two exogenous variables,

$$y_t = x_{1t}\beta_1 + x_{2t}\beta_2 + \epsilon_t. \tag{1}$$

Estimating the relationship while omitting x_{2t},

$$y_t = x_{1t}b_1 + e_t, \tag{2}$$

delivers a biased coefficient unless the regressors are orthogonal, $\mathbb{E}(b_1) \neq \beta_1$, and an upward biased residual variance, $s^2 > \sigma^2$. Both distortions matter in what follows, and the second is the one usually forgotten.

The same logic carries over to a VAR. Let

$$Y_t = B_1 Y_{t-1} + \cdots + B_p Y_{t-p} + u_t, \qquad \mathbb{E}[u_t u_t'] = \Sigma_u, \tag{3}$$

with $p \ll p^*$, where p^* denotes the true lag order. Omitting lags $p+1$ onwards biases the retained coefficients, because in any persistent system the omitted regressors are correlated with the retained ones, and inflates Σ_u, because variation that belongs to the systematic part of the process is pushed into the residual. Since the structural impulse responses are a function of both objects,

$$\text{IRF} = f\big(B(L), \Sigma_u\big), \tag{4}$$

they inherit both distortions. Note that this already implies something the horizon discussion below will sharpen: because Σ_u enters the identification, even the impact response at $h = 0$ is affected by lags the researcher chose not to include.

2.3 Two channels: pure truncation and identification bias

Ravenna (2007) separates the distortion into two channels. The first is *pure truncation bias*: a VAR(p) sets all coefficients beyond lag p to zero and therefore does not describe the true dynamics. The second is *identification bias*: where the autoregressive coefficients

enter the matrix mapping reduced form innovations into structural shocks, truncation corrupts that mapping as well. The identification bias arises *even when the identifying assumptions are correct*, and both results are established in population, so neither is an artefact of small samples.

Two implications matter here. First, Ravenna (2007) finds the identification channel to be the dominant one. Second, and more useful for the empirical work, the two channels do not treat all identification schemes alike. A short-run recursive scheme recovers the structural impact matrix from Σ_u alone, so it is exposed to truncation only through the residual covariance. A long-run scheme recovers it from the sum of the entire lag polynomial, $\sum_{i=0}^{\infty} \Psi_i$, an object in which truncation error accumulates rather than cancels, and which can be estimated very poorly even when the individual coefficients are estimated tightly. Ravenna (2007) documents parametrisations in which a VAR(p) approximates the infinite order representation well coefficient by coefficient and nonetheless tracks the long-run identified response badly.

This asymmetry generates a prediction that can be taken to the data: in a system where some shocks are identified with short-run restrictions and others with long-run restrictions, increasing p should move the second group considerably more than the first. Section **??** tests it.

2.4 The horizon structure of the bias

It is convenient to write the responses in moving average form. Stacking (3) in companion form, $X_t = AX_{t-1} + e_t$ with $X_t = (Y_t', \ldots, Y_{t-p+1}')'$, the response at horizon h to a structural shock is

$$\Psi_h = JA^h J' B_0^{-1}, \qquad J = [\, I_K \; 0 \; \cdots \; 0 \,], \tag{5}$$

where B_0^{-1} is the structural impact matrix delivered by the identification scheme. At every horizon the response is a deterministic function of the estimated coefficients. The distinction that matters is not whether Ψ_h is computed rather than estimated, since it is always computed, but what the data were allowed to say about it.

Horizons inside the lag structure, $h \leq p$. Here the regression included a regressor at lag h, so the sample covariance between Y_t and Y_{t-h} entered the estimation directly. This does not make the response unbiased. The estimated matrices $\hat{B}_1, \ldots, \hat{B}_p$ are not the first p coefficients of the true polynomial, for exactly the omitted variable reason set out above, and $\hat{\Sigma}_u$ is inflated. Truncation therefore biases the responses at every horizon, including impact.

It is worth being precise about a result that is sometimes read as reassurance. Ravenna (2007) shows that a correctly identified VAR(p) tracks the true response accurately up to lag p *in the case of zero identification bias*. That case requires the identification not to

depend on the lag polynomial. Where it does, and long-run restrictions are the leading example, the protection is absent. The technology shocks in the exercise that follows are identified exactly this way, so this is the relevant case rather than the reassuring one.

Horizons outside the lag structure, $h > p$. Here nothing in the data at lag h entered the estimation. The response is produced entirely by iterating a companion matrix that was fitted to shorter horizons. The VAR is no longer estimating the response, it is extrapolating it, and two things follow.

The point estimate is systematically attenuated. Beyond p the path in (5) is governed by the eigenvalues of $\hat{A}$, and a truncated system tends to understate the persistence of the process, because the persistence carried by the omitted lags has nowhere to go. The extrapolated tail therefore decays too quickly, and the longer the horizon the larger the shortfall. [**TBD:** Verificare la direzione del bias: in De Graeve and Westermark (2025) e Baumeister (2025) il risultato e' attenuazione, ma serve la citazione puntuale e va detto se vale in generale o nelle loro parametrizzazioni. Non sovraffermare.]

The confidence interval is simultaneously too narrow. Whether computed by the delta method or by bootstrap, the reported uncertainty at horizon h propagates the sampling variance of $\hat{B}$ through (5). It contains no allowance for the possibility that the AR(p) structure fails beyond lag p, because that possibility is not a parameter of the model being resampled. The interval measures uncertainty about the extrapolation, not uncertainty about whether extrapolating was legitimate.

The combination is the worst available. Beyond the truncation horizon the researcher obtains an estimate that is systematically too small, reported with a confidence band that does not know it is extrapolating. De Graeve and Westermark (2025) draw the practical conclusion, that a VAR(p) should not be used for structural inference at horizons $h > p$, and stress that this is in addition to, not instead of, the distortion at $h \leq p$.

[**TBD:** Figura schematica, una riga di orizzonti con $h \leq p$ ombreggiato come regione stimata e $h > p$ come regione estrapolata, sovrapposta a una IRF a gobba di ACEL. Renderebbe il punto immediato e costa poco. Valutare TikZ.]

2.5 Why longer lags help

The conventional objection to raising p is that the additional parameters make confidence intervals prohibitively wide. De Graeve and Westermark (2025) show that this argument is incomplete, and incomplete in a way that reverses its conclusion in realistic samples. Raising the lag order reduces misspecification, and reducing misspecification lowers the variance of the residuals as well as the bias of the coefficients. Two forces therefore act on the precision of the estimates and they pull in opposite directions: the parameter cost widens intervals, the variance reduction narrows them. The standard objection counts only

the first. In their simulations, at sample sizes comparable to those available in practice, the second frequently dominates, delivering lower bias, lower mean squared error, better coverage and, contrary to the received view, tighter intervals.

This is not an argument that parameters are free. It is an argument that the trade-off has two sides, and that the side which is usually ignored is the one that improves with p.

2.6 Information criteria answer a different question

If the lag order matters this much, the natural reflex is to let a selection criterion choose it. The reflex is misplaced. Akaike, Hannan–Quinn and Schwartz are constructed to optimise one step ahead forecast accuracy, an out of sample objective that penalises parametrisation. Structural inference is a different problem, and a criterion that trades bias against variance in the service of next quarter's forecast has no particular claim on the lag order required for an unbiased response at horizon twenty. De Graeve and Westermark (2025) put it directly: "information criteria focused purely on forecasting are poor guides for accurate structural inference". The point is not academic here, since ACEL report that the three criteria support $q = 2, 2, 1$ respectively, that is fewer lags than even their own baseline of four.

The symmetric warning applies to the in-sample diagnostics reported later in this thesis. Residual root mean squared error and generalised variance fall monotonically with p, but they fall in part mechanically, and a smaller residual variance is the common signature of reduced misspecification and of overfitting alike. Neither statistic can settle the question in sample.

The position taken here follows from this. I do not claim that $p = 12$ is the true lag order, and nothing in the argument requires it. The claim is that the structural conclusions depend on p, that this dependence is large, and that it should be reported as a matter of practice. This dissolves the model selection problem rather than pretending to solve it.

2.7 Choosing p: the horizon of the question, and the horizon of the estimator

Section 2.4 implies a rule that is more useful than any criterion: the lag order should be consistent with the horizons over which the analysis makes claims.

The horizon of the research question. If the object of interest is a long-run effect, then it lives at horizons where a short-lag VAR is extrapolating rather than estimating, and the finding will inherit the attenuation and the false precision described above. This is not hypothetical. Baumeister (2025) revisits the long-run real effects of monetary

policy reported by Jordà and Kozicki (2024) and shows that they are plausibly an artefact of truncation, disappearing once the lag structure is enriched. [**TBD:** Verificare l'affermazione esatta di Baumeister (2025): quali orizzonti, quali lag order, e se il risultato e' scomparsa o attenuazione. Serve la pagina.] The lesson generalises: a question about horizon h requires a lag order that reaches horizon h.

The horizon of the estimator. In impulse response matching this ceases to be a matter of judgement, because the horizons are built into the objective function. The estimator chooses θ to minimise

$$\big[\hat{\Psi}_{\mathcal{H}} - \Psi_{\mathcal{H}}(\theta)\big]' W \big[\hat{\Psi}_{\mathcal{H}} - \Psi_{\mathcal{H}}(\theta)\big], \tag{6}$$

where $\mathcal{H}$ is a fixed set of horizons fixed in advance by the researcher. Every element of $\hat{\Psi}_{\mathcal{H}}$ enters the criterion with a positive weight. Any element indexed by $h > p$ is not an estimate of the response at that horizon, it is the VAR's own extrapolation, and to that extent the DSGE is being fitted to the VAR's extrapolation rather than to the data.

The weighting matrix makes this worse rather than better. W is conventionally built from the estimated variances of $\hat{\Psi}$, so that more precisely estimated elements receive more weight. Since the reported variance at $h > p$ omits any allowance for the failure of the lag structure, those elements appear more precise than they are and are therefore weighted more heavily. The criterion leans hardest on the horizons that carry the least information. [**TBD:** Verificare dal replication package come e' costruito esattamente W in ACEL, e quali orizzonti compongono $\mathcal{H}$. Il testo sopra assume W diagonale con le varianze campionarie delle $\hat{\Psi}$, da confermare.]

The requirement that follows is a coherence condition on the estimator, not a robustness preference: p should be at least as large as the longest matched horizon. Under this reading even a researcher interested exclusively in short-run transmission must take a stand on p, because the estimator consumes the entire matched path regardless of which part of it motivated the exercise.

This condition is comprehensively violated in the exercise replicated here. ACEL match hump shaped responses that peak roughly one to two years after the shock and are matched out to around twenty quarters, with $p = 4$. The overwhelming majority of the matched horizons therefore lie outside the estimated lag structure. [**TBD:** Sostituire "around twenty quarters" con il numero esatto, e riportare la frazione di orizzonti matchati con $h > p$ ai tre lag order. E' una riga di numeri e regge tutta la sezione.]

3 Stochastic Singularity and the Absence of a VAR Representation

Chapter 2 argued that the exercise does not require the DSGE to admit an exact VAR(∞) representation. This chapter makes that argument concrete, because in the model being replicated the representation does not exist, and the reasons why are worth setting out plainly.

3.1 The standard licence, and where it fails

Fernández-Villaverde et al. (2007) provide the condition under which a state space system can be inverted into a VAR(∞) in the observables, namely that the eigenvalues of $A - BD^{-1}C$ lie inside the unit circle. The notation already carries the maintained assumption: D^{-1} must exist, so D must be square, so the model must have exactly as many structural shocks as observables.

This is not a technicality. When the number of shocks r falls short of the number of observables K, the system is *stochastically singular*. The spectral density of the observables has rank r, the Wold representation is $y_t = \Psi(L)\varepsilon_t$ with Ψ of dimension $K \times r$, and there is no non-singular K-dimensional white noise. The reduced form covariance matrix Σ_u is singular, Σ_u^{-1} does not exist, and a VAR(∞) representation in the standard sense is not defined. Ravenna (2007) states the practical consequence directly: singularity prevents likelihood estimation of the VAR, and recovering a non-singular representation requires dropping observables until the two counts coincide.

The VAR in Altig et al. (2011) confronts three structural shocks, monetary policy, neutral technology and capital embodied technology, with ten observable variables. The system is therefore singular, the invertibility condition is not merely violated but undefined, and there is no exact VAR(∞) representation of the model that the VAR is being used to estimate.

3.2 Why the exercise survives this

The reframing of Section 2.1 is what makes the exercise coherent. Impulse response matching does not require the DSGE to be the data generating process for all ten observables. It requires the VAR to be informative about the transmission of the three shocks being matched. Forni et al. (2016) note the cost of insisting otherwise: "if the number of variables in the VAR is smaller than the number of shocks in the DSGE, fundamentalness cannot hold (and the PMC is not even defined). Since modern DSGE models have several shocks, small VARs are automatically ruled out. This is unpleasant, since small VARs are useful tools, especially for short samples."

The consequence for this thesis is worth stating explicitly, since it explains what the exercise is and is not about. Singularity removes the possibility of defending the VAR as the exact reduced form of the model. What remains is the approximation argument, and the quality of the approximation is governed by the lag order. Singularity therefore does not compete with truncation as an explanation of the results reported later; it is the reason truncation is the only thing left to argue about.

3.3 How the literature has handled singularity

De Graeve and Westermark (2025) generate artificial data from three DSGE models, the real business cycle models of Chari et al. (2008) and Christiano et al. (2006) and the model of Smets and Wouters (2007), and estimate VARs on that data across short and long lag choices. Their design is explicit about the restriction just described: "invertibility is never a problem; all the models we consider are fundamental", and "all our experiments are based on two-shock models and two-variable VARs". The two real business cycle models satisfy this by construction. The Smets and Wouters model does not, since it carries seven structural shocks, and their solution is to reduce it, retaining "only monetary policy and preference shocks", until the number of shocks matches the number of variables in the VAR.

The direction of that adjustment is informative. The binding constraint is the size of the VAR laboratory, fixed at two variables throughout, so squareness is restored by discarding shocks rather than observables. Discarding observables is the alternative route, and is the one suggested by Fernández-Villaverde et al. (2007) and Ravenna (2007). In either direction the operation is the same: squareness is imposed so that truncation bias can be studied in isolation from singularity and invertibility. That is the right design for establishing the mechanism. It is not available to an applied researcher replicating an estimated model, and the present exercise does not attempt it.

3.4 The parametrisation frontier

There is a second respect in which this exercise departs from De Graeve and Westermark (2025), and it works against the hypothesis being tested. Because their data generating processes are two variable systems, their VARs remain cheaply parametrised even at long lag lengths, and raising p costs little in degrees of freedom. The ACEL VAR has no such room. With $K = 10$ variables, a constant and p lags, each equation carries $10p + 1$ coefficients, and the sample runs from 1959:Q2 to 2001:Q4, that is 171 quarterly observations before lags are consumed.

Table 1: Degrees of freedom across lag orders

	VAR $p = 4$	**VAR** $p = 8$	**VAR** $p = 12$
Usable observations T	167	163	159
Coefficients per equation k	41	81	121
Degrees of freedom $T - k$	126	82	38
Observations per parameter T/k	4.07	2.01	1.31
Coefficients, full system	410	810	1,210

Note: Each equation contains $10p + 1$ coefficients ($K = 10$ variables, p lags, one constant). The raw sample contains 171 quarterly observations (1959:Q2 to 2001:Q4), of which p are consumed as initial conditions. Ordinary least squares requires $10p + 1 < 171 - p$, that is $p \leq 15$; at $p = 15$ only five degrees of freedom per equation remain.

At $p = 12$ the system carries 1,210 coefficients, each equation is estimated on 159 observations with 121 regressors, and 38 degrees of freedom survive. Ordinary least squares requires $10p + 1 < 171 - p$, so no lag order above $p = 15$ is feasible at all. Tripling the lag length relative to the ACEL baseline is therefore not a conservative perturbation, it is very nearly the largest perturbation the data permit.

This makes the exercise a demanding test of the variance reduction mechanism of Section 2.5. In a two variable system the offset between parameter cost and reduced misspecification has ample room to operate. Here the parametrisation grows threefold in a system already large relative to the sample. If confidence intervals were going to explode anywhere, they would explode here.

3.5 They do not explode

Figure 1 plots the responses to a monetary policy shock at each lag order together with their own bootstrap confidence bands.

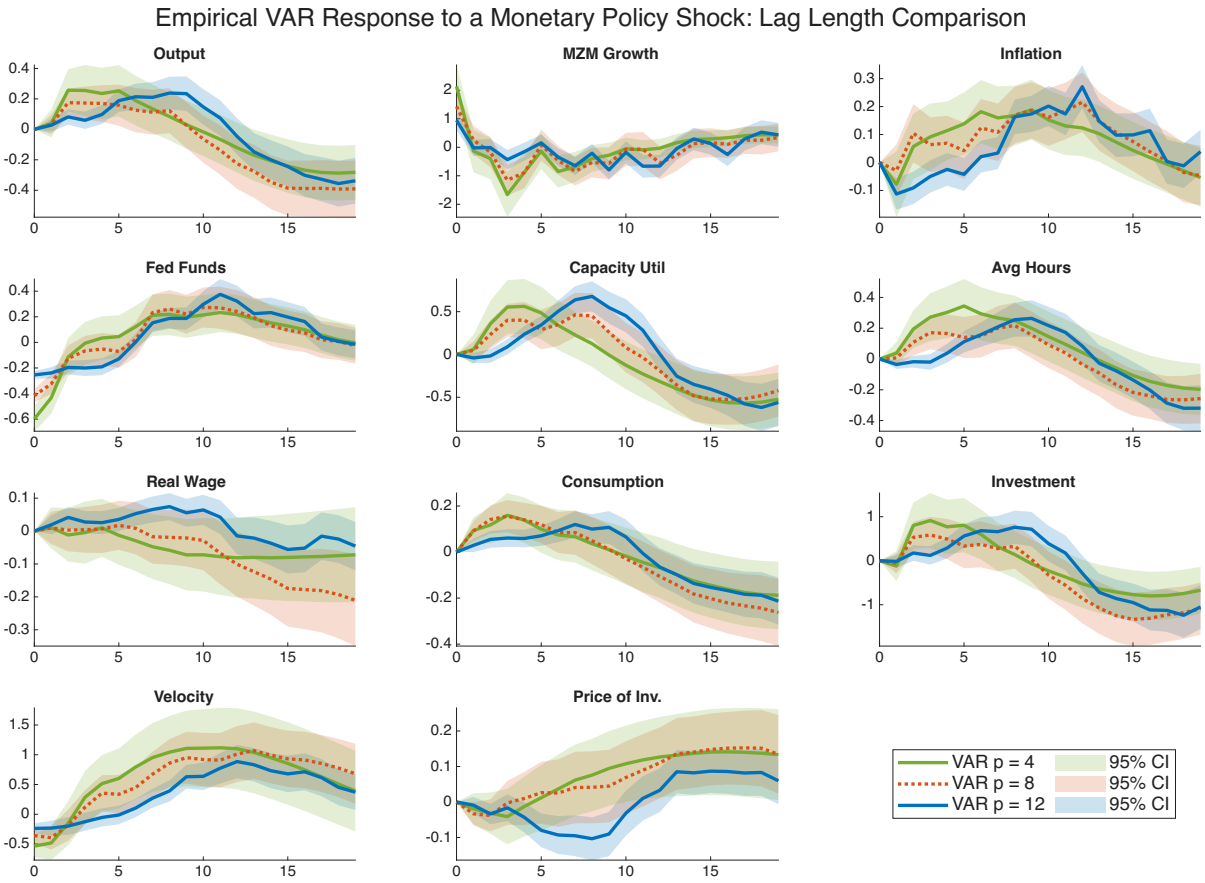


Figure 1: VAR impulse responses to a monetary policy shock across lag orders, each with its own 95 percent bootstrap confidence band.

The bands at $p = 12$ are not wider than those at $p = 4$, and for several variables they are visibly narrower, despite the threefold increase in parametrisation. The same pattern appears in the structural estimates, where the Phillips curve slope is estimated roughly six times more precisely at $p = 12$. The variance reduction mechanism therefore operates at the parametrisation frontier, which is a considerably more demanding test than the two variable systems in which it was first demonstrated.

[**TBD:** Quantificare invece di "visibly narrower": larghezza media della banda per variabile e lag order, e rapporto medio $p = 12$ contro $p = 4$. Una riga di numeri.]

Two features of the construction matter for reading the figure. The bands are bootstrap bands in which the entire VAR is re-estimated on each artificial sample, so the uncertainty induced by estimating 1,210 coefficients is propagated rather than conditioned away. Had the estimator been unstable at $p = 12$, the re-estimated responses would have varied wildly across draws and the bands would have widened accordingly. Draws yielding an explosive companion matrix are discarded and redrawn, so the bands condition on stationarity.

The obvious objection is that this is overfitting. A VAR with 121 regressors per equation fits the sample closely, the residuals shrink, and bands computed from those residuals shrink with them. The objection has to be taken seriously, because in sample it cannot be dismissed: reduced misspecification and overfitting share the same signature. What can be said is that the bootstrap re-estimation rules out the mechanical part of the concern, since parameter uncertainty is not being suppressed. What it cannot rule out is that the residual variance is itself understated, because the bootstrap resamples from the fitted residuals of the model under scrutiny. Distinguishing the two requires evidence that is not in sample, which is precisely what De Graeve and Westermark (2025) supply through coverage rates computed where the true responses are known.

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